

Lec 26 Entanglement, Disentanglement Latent Space, and Beta-VAE

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Transcript

[00:00] Hello everyone. Welcome to the lecture series on fundamentals of generative AI and the large language models. We are in the week four and this week four is all about the advanced variational autoencoders and practical applications. So far in the last week in the week three, we have seen in detailed about how VAE works and then we had a solved one numerical example and we had seen one of the most important concept called reparameterization trick. So with this we'll start with the week four and in the week four the following are the contents that we will be covering. So if you have seen in the week three, we had not seen any practical implementation.

[00:43] So this week four we will be starting with the hands-on implementation of VAE and then we will be seeing some of the important concepts such as entangled and disentangled latent space and we will see something called as beta VAE. Then we will move towards the conditional VAE. The when we are discussing the conditional VAE, we will also see what is the motivation. Why is that we have to go for conditional VAE and we will see the loss function of conditional VAE and there also we will see one

numerical example. And towards the end of this particular week in a theory session, we will see the next important concept called latent space interpolation. So and once we see the theory concepts of these, we will see a

[01:30] hands-on implementation on beta VAE and conditional VAE. So this is all the outline of this particular week four. So let's get started with in this particular session, we will see something called as entangle entangled and disentangled latent space and then we will connect something called as beta VAE. Okay. So although this This slide looks very heavy, but uh listen to this very carefully. It is very very simple concept. Now, what we are discussing is an entangled latent space. So, latent space is not something new to us. So, how is that we get the latent space?

[02:10] Now, uh we are uh passing There is some input. That input is a passed to the hidden representation. Here, hidden representation has a totally eight neurons. Then, is it not H1 is uh the combination of all this input X_1 to X_n ? Similar way, uh all the uh hidden neurons will have the combination of all of these inputs. Okay. What is that we are discussing? We are discussing VAE. Now, in VAE, how is that we are going to get the Z ? Z is obtained using reparameterization trick.

[02:49] Z is obtained uh using a formula. This is the formula which is used to get the Z . Which means uh the uh encoder also should produce us the mean and um uh variance, right? So, we have uh from um H , we are going to get this mean and then the variance. Now, how is that you will be establishing the connection? Is it not the connections are established in this way? Right? So, now, once we establish a connection like this, we have already seen in our previous uh week three last session where we have solved the numerical example. So, uh recall that particular concept. We have now obtained the mean and then the uh log variance.

[03:30] From log variance, you'll convert into variance, from variance to um you know, standard deviation. Okay. So, once this is uh done, is it not Z is now sampled using this particular reparameterization trick formula? Now, uh in this, what are we understanding? Now, when I get that Z , uh that Z is a combination of many of the important or many of these inputs. So, can I say because this mean whatever we have obtained, this mean has the combination of all of these. And all of this is the combination of this entire.

[04:09] So, now can we call Z as a latent space? Yes. Or a latent variable? Yes. So, now if I take a Z_1 , the latent representation Z_1 . Z_1 is a combination of all of the you know important variations. Then Z_4 to the Z_4 . Z_4 is a latent representation, which is the combination of all of those input. Okay. So, I hope you have understood this particular concept. Now, when we say Z_1 to Z_4 , what is the meaning of this? The meaning of this is let the input be of any dimension. Let the input be of any size. But in that input, there will be few important you know variations or important characteristic in that particular data.

[05:05] So, here Z1 to Z4 we are calling it as a latent representation. The role of a latent representation is to get that compressed representation of an input. Okay. Now, is this diagrammatic explanation is this clear? Input is positive and hidden representation because the hidden layer it is all of the neurons are connected to every of the hidden neuron. So, when I say H1, H1 is the combination of all this X1 to Xn. Similar way H8 is a combination of all of this. From H1 to H8, you are getting a mean and the variance. So, can I say that the mean whatever we have calculated, it is the combination of all of this? Yes? So, variance what we have obtained is once again is the combination of all of this.

[05:58] If mean and variance what we have obtained is the combination, from there when you get the Z1, do you agree that the Z1 is having the combination of all of those of features in a much compressed form? Agree to this, right? So, now let us see this is all about the latent space. Now, you are convinced that Z1 is a combination of many. Right. Now, let's go back and then see now what has been written in this particular slide. Let us consider there is an image of size 28 cross 28. We have considered an MNIST data set which is a standard handwritten digit, which is of size 28 cross 28. Now, I'm converting that into a single vector. So, single vector, it is of size 784.

[06:49] Okay. Now, if you look into this particular image, not all 784 pixels gives us some information. Uh there are only few important characteristics or important variations in this particular data. What is that variation? What is the thickness? Whether it is a thin or a thick? Then, whether it is a slant or whether it is straight? Then, what is the rotation, position? These are some of the important variations which are present in our data set.

[07:27] For example, one more if I have to give you an example which we'll be seeing in the next slide. When we have so many human beings you know, face data set. What will be the important variations? Is it not the important variations will be the skin, then the hair. It can be a smile or it can be the pose in which the person is giving or it can be the light. What is the amount of light? So, even though you have a data set, there'll be some of the important variations in that particular data set.

[08:02] Okay. So, I hope this concept is clear to us. Now, let us assume those important variations are these four. What are those four? This thickness, slant, rotation, and then the position. These are the four variations that we have. Now, what is that we are doing in VAE? We are discussing all about the VAE model. The encoder maps the input to whatever is the input, it is mapping it to these distributions. Yes. So, now suppose if you keep a latent space as four, if the dimension is four, then what is this four Z1, Z2, Z3, Z4 are the four latent representations.

[08:50] I hope this is clear to us. Now, since the hidden layer has eight neurons here, this eight is the combination of all X1 to X784. Agree to this? Similar way, H8 is a combination of all X1 to X784. If that is the case, do you agree to this concept? That is

Z1 to Z4, it is a mixture of it is a combination of all of those, that is the thickness, slant, rotation, and position. Do you agree to this? Yes? Because Z1, Z2, Z3, Z4 are the latent representations, which is responsible for capturing the important variations in the data.

[09:38] Now, if at all if there are four variations and this is Z1 to Z4 is a latent representation which is also having a four, then Z1 is just not one. It is a combination of all of those. Agree to this? Why it is combination of all of those? Because H1 to H8 is a combination. H1 to H8 when it is passed to mean and variance, there also it is a combination. From there when you get a Z1, it is once again the combination of all of these of four.

[10:11] This is important concept. You need to understand this. So, this is a called an entangled latent representation. What is entangled latent representation is Now, changing one latent variable will change the multiple properties together. So, what is what is the meaning of it? Now, if you take a Z1, Z1 has some portion of thickness. It has some portion of you know, the next slant. It has some portion of rotation. And it has some combination of a position.

[10:55] Now, what is entangled latent representation is you look at this a Z1. Z1 is a combination of all of these you know, variations. Now, if I want to do some modification to this a Z1, I want to change some value. Some modification you want to do to this a Z1, then if you do the modification to Z1, it is having an impact on this. It is having an impact on all of these. So, changing one latent variable may change multiple properties together.

[11:29] Okay. Suppose if this concept is not very clear, probably when we see what is disentangled, you may understand this. From this particular slide, understand only one concept called entangled latent space. What is entangled latent space? Z1 to Z4 is the latent size. Now, each of these latent variable is the combination of many of such important variations. So, by doing one small modification, you will have an impact on all of those.

[12:05] This is called entangled. Now, further if you have to understand, this is look at this particular diagram. Now, when you see this first this is all the human face. In this, what are important variations? The important variations are hair, then what is the amount of lightning on the face, skin, smile, and pose. These are the different important variations in this particular data set. Now, can we have a model or how about this architecture? Now, if you look into the first first set of images, the person is the same. Here, the person is the same. Person is the same here also. Here, the person is the same. But, what has a changed here? If something has a change, it is only the hair.

[13:02] Now, if if at all if we can capture if I if we can do a modification only with respect to hair, is it not we are getting multiple variations of same data? And each of

this variation is different from each other. Is it not we are generating more such data? Now, look at this particular smile. When you see this, here the person is the same, but look at the first image, look at the second, third, and the fourth. As uh and the difference is only change in the smile. By changing only one uh parameter, you are able to generate a different variation. Similar way, look at the pose pose. Now, when you look into this, here once again the person is the same, but uh the this particular variation, if you can able to

[14:00] modify only this, keeping rest of the parameters the same, is it not we are getting a different version? So, this concept is um good, right? It is more convenient to get more amount of data to generate a different variations in the data, right? So, now, what is this concept is all about? You have an hidden representation, the you have a latent representation. What is this latent representation? Here, it has to capture the important variations. Some time back we have seen that those hidden representation is capturing, but there is a combination of many of these important variations. Z1 is having all of these variations in different combination. Z2 is also having all of those important variations. Instead of

[14:56] that, can we do something like this? If at all if I have an latent representation size as a five, and if here the important variations here are five, can I make the model where each of the latent representation is a primarily focusing only on one variation? Which means the latent representation Z1 is primarily focusing on hair. Z2 is a primarily focusing on light, Z3 on skin, Z4 on a smile, and a Z 5 as the pose. If you can able to achieve this, if we can make each of the latent representation to capture only one of the important characteristic of this variation, then what is that we are going to achieve? So, here if at all if you want to do this, then each latent variable mainly controls only one factor

[15:53] of variation. True? Now, by doing this, now when we change only Z1, look at this. Now, when we modify only Z1, which means only hair is getting modified, then rest of the concept is a remain the same. So, therefore, if you change only Z1, mainly the hair-related features should change. Other factors, such as the lighting, skin, smile, and then the last characteristic, everything should remain unchanged, mostly unchanged. If you can able to do this, then what are we achieving? Is it not we are manipulating a specific latent variable? By manipulating only a specific latent variable, the model can generate the variations in the image.

[16:45] Now, if I'm changing only the hair in this particular one, is it not I'm able to control I'm able to generate the different variations in the image? And is it not it is going in a much controlled way? Um is this concept okay? All right, how to do we will learn that. But is this concept of Did you get this concept? Now, if you can anyhow latent representation role is to capture the important variations in the data in a lower dimensional space.

[17:21] Now, anyhow it is capturing, can we enforce, can we ensure that each of the latent variable should be responsible for capturing only one of the variant, one of the important characteristic. If we can achieve that, then that is called disentangled latent space. What is a disentangled latent space? It is a latent space where the primary aim is to learn the structured representation. What is a structured representation here? Where each of the latent dimension is focusing only on one factor of variation. Let any number of variations be there, but each of the latent vector is responsible for capturing only one important variation.

[18:07] So, I've given the same example here. Suppose if you consider the five latent size as a five, then we have a totally five different you know, vectors, Z_1 , Z_2 , Z_3 , Z_4 , and Z_5 . So, our primary aim is Z_1 should focus only on the hair. Z_2 should be focusing only on lighting. Z_3 should be focusing on skin tone. Z_4 on a smile and Z_5 on the pose in which the face is. So, if these are the meaningful factors in the image, then we want each of the latent vector to focus only on capturing one of that important.

[18:49] So, what is our ultimate aim? Ultimate Is it not the ultimate aim is by changing only one, we are keeping the rest of it same. So, changing one latent vector changes only one particular that factor, which means if you are increasing Z_1 or if you are changing a Z_1 , only hair related variation changes. Similar way when we change only Z_2 , only lighting should change, the rest of it should remain the same. When we take Z_3 , skin tone should change, but the rest of it should remain.

[19:26] That is the reason if you go back and then look into this diagram. Uh, so if you change if you are looking into there is only a smile which we have considered and only this particular you know Z is being modified or it is being changed. By changing that rest all of the factors has remained the same. Only the smile has been modified. This is a called disentangled latent space. So where you are you know if you change only one latent variable then mainly only one property of the generated image should change and the rest of the features should remain the same. So if you look into this so mainly only one property of generated image should change. So all these are the generated images and only one parameter

[20:26] has been changed. What is that parameter? It is only the hair which is being changed. And rest all of the properties has remained the same. So do you agree to this particular concept? This is a called an disentangled latent space. So by doing this we are going to have an interpretable and controlled latent space and this is called disentangled latent representation. I hope this concept is clear to you. Now how do we achieve this? Okay, latent representation Z so far whatever we have seen it is an entangled. So, if that is the case, how do we achieve with the disentanglement? So, in a standard VAE, the loss function is this. We have a first term as reconstruction, and the second is the KL divergence.

[21:22] To this particular loss, we are adding one regularization parameter called

beta. So, adding the adding a weight beta to a KL divergence term. So, this is what is the beta VAE, where the only difference here is there is one regularization parameter that is a beta is being added to the KL divergence. Now, if that is the case, let us understand how it is going to achieve this. Okay. Now, what is the first term here? The first term is a reconstruction. What is the reconstruction primarily ensures that it ensures that the decoder reconstruct the input. So, which means whatever the reconstructed data, that should be approximately closer to the original input. Which means the latent representation is Z .

[22:14] How do you get the decoder How decoder will reconstruct? Is it not Z should be passed as an input to the decoder? If Z is very good, then definitely decoder can reconstruct the input reconstruct the data well. So, which means store enough information in a Z so that the decoder can able to reconstruct it. Okay. Let's now look into the second term. What is the second term here? KL divergence. What is KL divergence says?

[22:44] It always encourages the encoder distribution should remain closer to the prior distribution. That is what the second term KL divergence says. So, now which means do not store the information in an arbitrary way. Why? Because if you store it in an arbitrary way, then definitely the you know, the distribution the difference between these two distribution will be very high. So, one term is saying that store enough information in Z so that I can able to reconstruct well.

[23:17] The second term is saying do not to store the information in an arbitrary way. So, which means there should be a good balance here. And that balance is there has to be a good reconstruction, but at the same time you know, we should have an a regular latent space. Okay. Now, you can ask this particular question. What is the difference? Even in standard VAE also the loss we have seen that reconstruction loss and KL divergence.

[23:47] So, all this is doing even in standard VAE also, then in what way it is a different is look at the loss function very clearly here. What is if nothing is there, can we assume this as one? If it is one, can I say that the beta term here is one? Right? So, whatever is the beta value in a standard VAE beta value is one. So, which means model can allow the latent representation to have some mixture. What is that mixture?

[24:20] We have seen sometime back the combination of many of those important variations. As long as it is helping you for a good reconstruction, go ahead and then have some combination the latent representation can have some combination. Which means it is a primarily not primarily it is we can say that it is an entangled latent representation. So, which means it does not guarantee the disentanglement. In a standard VAE, there is no guarantee that there is a disentanglement.

[24:53] What do you mean by disentanglement? Each of the latent representation

should capture only one variation. Standard VAE beta value is one. Because beta value is one, there is no guarantee that the scale divergence, you know, when beta value is one, there is no guarantee that the latent representation is disentangled. You can have the combination. You will have the mixture of the important variations which are represented in the latent representation. Therefore, it is an entangled latent representation, but it does not guarantee the disentanglement.

[25:36] If you have to achieve disentanglement, what is the only possibility? What should be the beta value? The beta value should be greater than one, right? In beta VAE, beta value will be always greater than one. Now, just imagine what will happen if beta value is greater than one. Is it not more importance is given to the KL divergence here? When more importance is given to this, is it not little more extra pressure is given to the model?

[26:05] What is that pressure? The pressure is definitely the encoder distribution should follow the prior distribution. Is it not this the meaning that we are getting? Okay. So, when KL value when beta value is greater than one, KL term is given more importance, which means a larger KL weights, you know, there is more pressure that is being put. And what is that pressure here? Encoder distribution should stay closer to prior distribution. Right? So, what do you what do you mean by this? It strongly encourages that the mean value should be closer to zero and then the variances should be closer to one.

[26:50] So, this restricts the model of from freely storing mixed information in all the latent variable. So, this is the place where you are restricting the model to store mixed information in the latent variable. So, this is the beta VAE where the loss function will remain same exactly as standard VAE, but there is one regularized term called beta and beta value should be more. And what is more should be greater than one. When it is greater than one, how is that we are achieving? There is a There is a little more stronger pressure on this particular term. What is that? Encoder distribution should stay closer to prior distribution.

[27:37] When you are enforcing that this enforce ensures that the model does not capture too much of mixed information at a latent variable. Right? So, now the the next concept here is So, just by having the beta term here, are we very sure that the disentanglement is achieved? There is no guaranteed that perfect disentanglement is achieved. So, you are encouraging the disentangled latent space by applying a stronger regularization, but however, there is no perfect There is no guaranteed that you will achieve with the disentanglement here.

[28:24] Now, so can you think? Okay, if disentanglement is not guaranteed, can I increase the beta value so that definitely disentanglement is achieved? Answer is no. If you If you make a beta as a too large value, then what is that you are doing it? You are compromising on the reconstruction quality. So, the reconstruction quality may

decrease. When reconstruction quality decreases, definitely the reconstruction loss increases here.

[28:57] So, this is what is an issue. Now, how is that we are you know little more let us have little more clarity on whatever is the theory concept that we have seen so far. So, let me summarize before we get into a numerical intuition or example. The whole idea here is you want something of this. Let's go back and then uh Yeah, our aim is if we have our model where each of the latent space captures the important variation and every of the vector is a responsible for capturing only one of the variation.

[29:42] If that is achieved by doing modification only with respect to Z_1 , you can able to generate a different variations. So, this is our requirement and this requirement is achieved by using the KL you know beta term is added to an KL divergence. Now, how beta is going to control here is if beta value is greater than one, the this entire KL term becomes more. When KL term becomes more, which means a stronger pressure is being put to the model that the both distributions should remain closer to each other.

[30:23] This is the restriction which we are putting. This restriction ensures the model does not capture that is the hidden representation or the latent representation will not have too much of combination, but it tries to go for an you know single important Now, let us look into and from this what is that we have understood? Although, you know, beta VAE does not guarantee the perfect disentanglement. If you try to increase the beta value to a larger one, then the reconstruction quality will be compromised. Okay. Now, let us take one small example so that you probably you'll be able to connect with whatever the theory that has been explained.

[31:11] Now, let us assume our latent space size is two, just for an understanding. If it is two, then is it not Z_1 Z_2 ? How is that you get Z_1 Z_2 ? Is it not mean one plus standard deviation one and then the epsilon one? Z_2 , [snorts] is it not Z_2 is obtained from this particular formula? And epsilon one and two are sampled from normal distribution. Okay. Now, what is the possibility? The possibilities are either entanglement or disentanglement. Now, let us assume ideal case where it is disentangled latent space.

[31:49] If it is a disentangled latent space, which means Z_1 is responsible only for one factor, Z_2 is responsible only for one factor. So, can I say Z_1 , for example, Z_1 is responsible for thickness, Z_2 is responsible for slant? Now, how do you get this Z_1 and Z_2 ? Is it not you are getting one and Z_2 from mean one and then standard deviation one? And Z_2 is from these two, right? So, in an other way, is it not if Z_1 you are calling it as a disentangled, which means that mean one is mainly primarily focusing on thickness?

[32:33] Then only Z_1 can be only on thickness, right? So, can we say Z_1 is a focusing if it is focusing on thickness and if Z_1 is obtained from mean one, then mean one is also focusing on thickness, right? So, mean one mainly encodes the thickness and

Z2 carries the slant information. Therefore, mean two is also focusing on this. Okay. So, what is the hidden intuition? Mean one is capturing only little information. What is that little information? It is capturing only thickness. Mean two is capturing only little information, which means it is capturing only thickness.

[33:19] That is a slant. So, if it is capturing small information, is it not the mean values should be small? Can you understand in this way? So, if Z1 should focus only on thickness, Z1 is obtained from mean one, then mean one should be capturing only a small information. What do you mean by small information? Only one information. So, here mean one and mean two values are small. Why small? Because it is focusing on capturing only that particular information. For simplicity, we have assumed the second that is variance as one.

[34:00] Just for simplicity for calculation. So, when you calculate this, is it not this is a KL divergence in a closed or form? So, when since this variance value is one, so substitute the values in this. This is one log one minus one. Log one is a zero. So, one minus zero minus one, that entire terms gets cancelled. So, KL divergence, you are left out with only the mean value. » [snorts] » Since the latent representation is two, therefore we are taking both mean one and the mean two. So, here the value is 0.4 and 0.2 just for an example.

[34:40] Substitute the values. KL divergence value what you have got is very less. So, from KL divergence value when you have got very less, what is the ideal meaning of this? Can you go back and then understand what is the meaning of the KL divergence value is less? It means encoder mean or encoder distribution distribution is very much closer to the prior distribution. Is it not the distribution is closer to the prior distribution? So, if it is very much closer to the prior distribution, when this is possible? Only when the mean values are less. If the mean value is less, is it not that it has captured the less information? Are you able to backtrack and then do the reverse engineering explanation here? So, I'm

[35:31] back once again. So, here we need to understand how it is ensuring that Z is focusing only on the one variation. So, if Z is focusing only on one variation, which is on thickness, is it not that thickness Z1 is obtained from mean one. If Z1 has to focus only on this, is it not mean one also should focus only on this? Mainly encodes the thickness. If that is the case, is it not it is carrying only one information? If it is carrying one information, can we say that the value is small?

[36:12] If the value is small, then what is that it is giving us the information? Overall KL value is less. If overall KL value is less, what is that it means? That encoder is closely matching with the prior distribution which means the mean value is closer to zero. If the mean value is a closer to zero, which means that can we say that the these mean values are taking only less information. If it is less information, is it not that Z is focusing only on one parameter?

[36:47] I hope this is clear. Let's take one more example where it is entangled space. Now, what do you mean by entangled? Entangled means since the latent representation size is two, so $Z_1 Z_2$. Z_1 should be combination of both. Z_2 should be combination of both. Okay, if Z_1 is a combination of both, then mean one also should be the combination of both. If Z_2 is a combination of both, then mean two also should be the combination of both. If that is the case, definitely the value will be higher.

[37:24] Look at this. The value will be higher. If the value is higher, can we say what will be our KL divergence value? Here the KL divergence value, can you see? The value is high where you're getting KL value as 0.50. So here once again, mean value what we have considered is 0.8 and 0.6. Substitute the values back into this equation. So taking the square of it, it is 0.64 and 0.6 square is 0.36. Add up these two and then the KL divergence value is large.

[38:01] When KL divergence value is large, what is that we are understanding? Here it is 50% of it. So what is that we are understanding? Is it not that the encoder mean is away from the or far from not away, far from the prior distribution? Now, how can you say it is a far from the prior distribution? If it is closer or if it is nearer, then the mean should be closer to zero. But, do we have our mean closer to zero? No. The mean values are not closer to zero. If the mean values are not closer to zero, it means mean value is more.

[38:38] What What is the meaning of that mean value is more? It has captured more information. So, here the distribution is far from the prior and the encoder is using a more uh uh is using a more latent capacity to store the information. Now, let us uh take one more example and so that you know you'll be um further convinced. Here, further mean has increased. If mean has increased, then what what is the meaning of it? Uh the encoder distribution is a far from the um you know prior distribution. If you look into this particular example, here it is 1.5. When you square it, it is 2.25.

[39:19] Then, the second is 0.5. So, here the KL value is very large. When KL value is very large, what is the meaning of it? Encoder distribution is far from the prior distribution. Uh what is again the other meaning? The mean of it is a far from the normal distribution zero. If it is a far from normal distribution, look at the value is very high. If the value is very high, it means that it is capturing more information. If it is capturing a more information, definitely from there Z_1 , whatever you get, that is the you know, uh combined one. So, this is an entangled.

[39:59] Right? So, I hope this is uh clear to you. Now, I have one particular problem here. Now, what is this? Suppose you have an image where in the image you have you know less number of important variations and you have almost uh you know the latent dimension similar to that which means whatever example that we have seen so far that example had four important variations and the latent space also had a set. So because of latent variable Z also had four. So because of this there can be a possibility

that entanglement is disentanglement is possible because it is one-to-one mapping. But can we consider this a scenario where in the image the number of variations are large. There are many such important

[40:57] variations. For example, look at this you have the important variation. There are six such important variations. Now so far in our understanding we had to taken equal number. But what now you have in a latent space which is smaller than or lesser latent space size is lesser than these important variations. So here the variations are I mean latent size is only three. What is the possibility? There are two possibilities. What is the first possible? Allow the entanglement to happen. That is one approach. So the first approach is allow the entanglement to happen. If you allow the entanglement to happen, then then only it can able to capture all of those information.

[41:50] If it captures all of those information, then only the reconstruction will be proper. So if you allow the entanglement to happen here, this is the only solution that we are able to see at this point of time. Otherwise, the number of variations are more. Z is a small. If you allow the disentanglement, then it is going to capture only few. Remaining are left out. So, if you allow the entanglement to happen, then is it not a Z1 is a combination of all of those uh features? Z2 is the combination, Z3 is the combination. All of the features have been captured.

[42:26] If all of the features have been captured, then what is very good? Reconstruction is good. So, if this is the case, since the model is uh storing more information, reconstruction will be better. Therefore, the reconstruction loss will be lower. But, what is the problem? Some time back in our example, we have seen if it has captured more information, what what will increase? The mean value will increase. When the mean value increases, it is definitely far from the prior distribution, which means KL divergence value. What what is that? See, because the latent variable is carrying a more information, then the encoder distribution may vary far from the you know, prior distribution. If that is the case, what is going to

[43:12] happen? Is it not the KL divergence value will increases? We have seen in two examples in the previous slide. KL divergence value increases. If a KL divergence value has increased, which means what? So, if a disentanglement should happen, then definitely beta value should be greater than one. So, in this particular case, what will be the beta value? Beta value will be small, will be very low, because there is no disentanglement. It is entangled. When it is entangled, beta value will be smaller.

[43:45] Is this example is clear to you? There is one scenario where the number of input features, important features are greater than the latent size. If that is the case, one scenario is entanglement. One solution is entanglement. If [snorts] you allow the entanglement to happen, then KL divergence increases. So, then what is that you have to do? You have to go for disentanglement. You have to encourage to go for

disentangled latent space. Now, what is the problem here? If you go for disentangled latent space, now there are six such important features and the latent size is only three. If that is the case, is it not only among this what is the most important? It is going to capture the uh those features and the remaining

[44:37] are not at all been used or not at all been captured. If it is not captured, is it not the Z is very weak? If a Z is weak, if Z is passed to a decoder, then reconstruction will be very poor. In this particular scenario, what is going to happen? Is it not the reconstruction loss increases? Definitely, reconstruction loss increases. Why? We have bad Z. When Z is poor, when that Z is passed to an decoder, reconstruction will be very reconstruction loss will be very high.

[45:09] So, now because it is a disentanglement, beta value is greater than one. If the beta value is greater than one, what is that we are ensuring? That the mean value should be closer to our prior distribution. That means the KL divergence value will be very less. So, in this particular scenario, what is going to happen? That the KL divergence value may decrease. So, here there is an high beta value in this particular scenario.

[45:38] Because there is a high beta value, more emphasis is given. Because it is more emphasis, so definitely this distribution should stay closer to this. Which means that the mean value will be closer to zero. When mean value is closer to zero, when you calculate the KL divergence term, the KL divergence term decreases. We thus can we ensure that the has focused only on a few. Reconstruction loss has increased, KL divergence term has decreased. Before in the previous example, what has happened? Reconstruction loss has you know, decreased, but KL divergence term had increased. So, in this both are not the solution for a given scenario. Then, what is the solution? In but in such particular case, one of the you know, possible

[46:28] solution is increase the latent space. Now, how do you ensure that how much to increase? It is just an a trial and error. It is a hyper parameter. Increase the latent space size. So, by increasing the latent space, we will be able to capture more features. So, you you think of disentanglement. So, more such a features will be captured when more such reconstruction loss will decrease. When reconstruction loss decreases, anyhow KL divergence term is also decreased. Right? So, this is what is you know, one possible solution.

[47:06] But here, this requires you know, you have to go for it's an hyper parameter tuning. So, what is the second possible solution? So, second possible So, here what exactly by increasing the latent space, what it is saying? So, this gives the model more capacity to capture more meaningful factors separately. So, more latent space means more information is captured. Therefore, there is a better reconstruction and there is a lower reconstruction loss. So, second possible solution is tune that beta value. So, how is that you tune the beta value? We have seen sometime back if the

beta value is very large, then the quality reconstruction is very poor.

[47:52] So, if the beta value is very large, then there is more enforce on KL term, which means the mean should remain more closer. Then which means we get a Z as very weak Z. If that Z is very weak, reconstruction loss increases. So this also cannot be the solution. You cannot have beta value very large. What is the solution here? The one solution is to choose a moderate beta, but not very high. Once again, here it's an hyper parameter. So keep [snorts] changing the values, you'll be able to get a balanced between reconstruction and disentanglement.

[48:30] Next is use beta annealing. So the third solution is beta annealing. What is beta annealing? Can we start beta value? So go back to the reconstruction loss and then think, can I have beta value as zero? So when we have beta value as zero, entirely second term becomes a zero. So which means you are enforcing the model to learn only focus only on reconstruction first. Get a good quality reconstruction. Okay? So instead of using a very high beta value, start choosing a beta with a very smaller value and gradually you increase it. So how is that you'll be doing? So this allows the model to first learn the good reconstruction and it gradually improves the latent structure.

[49:19] » [snorts] » So beta annealing here is beta can start with zero or a very small value during the initial stage of training. And when the beta value is very small or zero, is it not the second term ideally becomes zero? Then is it not more importance is given only to the first term? First term is the reconstruction. So you're first allowing the model to focus on good reconstruction. Then at this particular stage, KL value will be very small or you know, the impact will will very less. So the model focuses only on a learning a good reconstruction first.

[49:55] Then gradually increases increase the beta value. By doing this, then the KL term becomes active and it starts encouraging the disentanglement. So, this is all about this particular concept. With this, we have come to an end of this particular session. In the coming session, we will be looking into an conditional VAE. Thank you.